

06 BINOMIAL AND POISSON (ppt3, 28 pages, lectures 12-13)

PART A: BINOMIAL

1. The derivation from 5 coin tosses (p003-p007)

QUESTION: $P(\text{exactly 3 heads in 5 tosses})?$

one arrangement: HHHTT

$$P = (1/2)(1/2)(1/2)(1/2)(1/2) = (1/2)^3 (1/2)^2$$

another: THHHT

$$P = (1/2)(1/2)(1/2)(1/2)(1/2) = (1/2)^3 (1/2)^2$$

EVERY arrangement with 3 H and 2 T has the SAME probability.

how many arrangements? 5 positions, choose 3 for H

$$C(5,3) = 5!/(3! 2!) = 120/(6 \times 2) = 10$$

the 10 arrangements, all equally likely:

THHHT	HHHTT	TTHHH	HTTHH	HHTTH
HTHHT	HTHTH	HTHTH	HHHTT	THHTH

THEREFORE:

$$\begin{aligned} P(3 \text{ heads and 2 tails}) &= C(5,3) \times P(H)^3 \times P(T)^2 \\ &= 10 \times (1/2)^5 \\ &= 10/32 = 0.3125 = 31.25\% \end{aligned}$$

THE STRUCTURE, IN ONE PICTURE:

[number of arrangements]	x	[probability of one]
$C(n,x)$		$p^x q^{(n-x)}$

2. The setup conditions (p008-p011)

a random experiment performed repeatedly

"success" = the event occurs, "failure" = it does not

PHYSICAL CONDITIONS FOR A BINOMIAL

1. each trial gives TWO mutually disjoint outcomes
2. n, the number of trials, is FINITE
3. the trials are INDEPENDENT
4. p, the success probability, is CONSTANT

$$q = 1 - p \quad (\text{failure probability})$$

3. The definition (p009-p010)

$$\begin{aligned} P(X = x) &= C(n,x) p^x q^{(n-x)}, & x = 0,1,2,\dots,n \\ P(X = x) &= 0 & \text{otherwise} \end{aligned}$$

n and p are the PARAMETERS. n is also called the DEGREE.

NOTATION: $X \sim B(n, p)$

4. Mean and variance

$$\begin{aligned} E(X) &= np \\ \text{Var}(X) &= npq \\ \text{SD} &= \sqrt{npq} \end{aligned}$$

5. WORKED: pens (p013-p016) - the flagship example

10% of pens are defective. A box has 12 pens.

X = number of defective pens in the box.
 $n = 12, p = 0.1, q = 0.9 \quad X \sim B(12, 0.1)$

(i) EXACTLY 2:

$$\begin{aligned} P(X=2) &= C(12,2) (0.1)^2 (0.9)^{10} \\ &= 66 \times 0.01 \times 0.3487 \\ &= 0.2301 \end{aligned}$$

(ii) AT LEAST 2 (use the COMPLEMENT):

$$P(X \geq 2) = 1 - P(X < 2) = 1 - [P(X=0) + P(X=1)]$$

$$P(X=0) = C(12,0) (0.1)^0 (0.9)^{12} = 0.2824$$

$$P(X=1) = C(12,1) (0.1)^1 (0.9)^{11} = 12 \times 0.1 \times 0.3138 = 0.3766$$

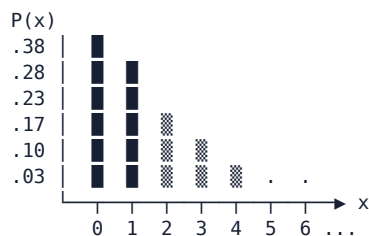
$$P(X \geq 2) = 1 - [0.2824 + 0.3766] = 1 - 0.659 = 0.341$$

(iii) NONE:

$$P(X=0) = 0.2824$$

THE "AT LEAST 2" MOVE IS THE WHOLE POINT OF THIS EXAMPLE.
 Summing $P(2)+P(3)+\dots+P(12)$ is 11 terms.
 Complementing is 2 terms. ALWAYS complement for "at least".

the pmf picture for $B(12, 0.1)$:



$P_0 = 0.28, P_1 = 0.38$ <- note $P_1 > P_0$ here ($p=0.1, n=12$, mode at $x=1$)

6. WORKED: irregular die (p017-p018) - the parameter recovery

A six-faced die. $P(5 \text{ evens in } 10 \text{ throws}) = 2 \times P(4 \text{ evens in } 10 \text{ throws})$.
 In 10,000 sets of 10 throws, expected number with NO even number?

set the condition:

$$C(10,5) p^5 q^5 = 2 \times C(10,4) p^4 q^6$$

$$252 p^5 q^5 = 2 \times 210 p^4 q^6$$

$$252 p = 420 q \quad (\text{divide by } p^4 q^5)$$

$$3p = 5q = 5(1-p)$$

$$3p = 5 - 5p$$

$$8p = 5$$

$$p = 5/8, \quad q = 3/8$$

then:

$$P(X=0) = C(10,0) (5/8)^0 (3/8)^{10} = (3/8)^{10} = 0.00005499$$

$$\begin{aligned} \text{expected number out of } 10,000 \\ = 10000 \times 0.00005499 = 0.5499 \end{aligned}$$

TWO SKILLS IN ONE QUESTION:

1. recover p from a ratio $P(X=k+1) = c P(X=k)$
 (the C's and the powers of q cancel, leaving a linear eq)
2. EXPECTED COUNT = $N \times P$ (not P alone!)
 0.5499, NOT 0.00005499

ERRATA 2: the slide prints 0.549; the exact value is 0.5499.

PART B: POISSON

7. What it models (p019-p021)

Poisson models the number of "RARE EVENTS" occurring randomly in a fixed interval.

the slide's seven instances:

1. deaths from a disease (heart attack, cancer, snake bite)
2. suicides reported in a city
3. defective material in a packing
4. faulty blades in a packet of 100
5. air accidents in a unit of time
6. printing mistakes per page of a book
7. cars passing a crossing per minute (busy hours)

8. Poisson as the limit of binomial (p022)

under these three conditions:

1. $n \rightarrow \infty$ indefinitely large
2. $p \rightarrow 0$ indefinitely small
3. $np = \lambda$, finite $p = \lambda/n$, $q = 1 - \lambda/n$

$$C(n, x) p^x q^{n-x} \xrightarrow{\text{limit}} e^{-\lambda} \lambda^x / x! \quad \text{with } \lambda = np$$

"USE THE APPROXIMATION" IN A QUESTION MEANS:
Poisson with $\lambda = np$, NOT the exact binomial.
e.g. 5000 men, $p=0.001 \rightarrow \lambda = 5$

9. The definition (p023-p025)

$$P(X = x) = e^{-\lambda} \lambda^x / x!, \quad x = 0, 1, 2, \dots, \lambda > 0$$

$$P(X = x) = 0 \quad \text{otherwise}$$

λ is the PARAMETER. NOTATION: $X \sim P(\lambda)$

$$E(X) = \lambda \quad \text{Var}(X) = \lambda \quad \leftarrow \text{MEAN EQUALS VARIANCE}$$

A POISSON VARIABLE HAS MEAN = VARIANCE
Equality is necessary for Poisson, but it does not prove that
an otherwise unspecified distribution is Poisson. Use the
model assumptions or a stated Poisson family as well.

10. WORKED: parameter from a ratio (p025)

$P(X=1) = 0.2 P(X=2)$. Find $P(X=0)$.

$$\frac{e^{-\lambda} \lambda^1}{1!} = 0.2 \frac{e^{-\lambda} \lambda^2}{2!}$$

$$\lambda = 0.2 \lambda^2 / 2 = 0.1 \lambda^2$$

$$1 = 0.1 \lambda \quad (\text{divide by } \lambda, \lambda > 0)$$

$$\lambda = 10$$

then:

$$P(X=0) = e^{-10} 10^0 / 0! = e^{-10} = 0.0000454$$

the cancellation picture:

$$\begin{array}{ccc} \underbrace{e^{-\lambda} \lambda}_\text{cancels} & = & 0.2 \times \underbrace{e^{-\lambda} \lambda^2}_\text{cancels} \\ \lambda & = & 0.2 \lambda^2 / 2 \\ \longrightarrow & & \lambda = 10 \end{array}$$

11. WORKED: telephone calls, THE NESTING (p026)

calls arrive at $\lambda = 2$ per minute.

$P(\text{exactly two calls in each of the first 5 minutes})?$

STEP 1: Poisson within ONE minute

$$P(N=2) = e^{-2} 2^2 / 2! = 2e^{-2} = 0.270671$$

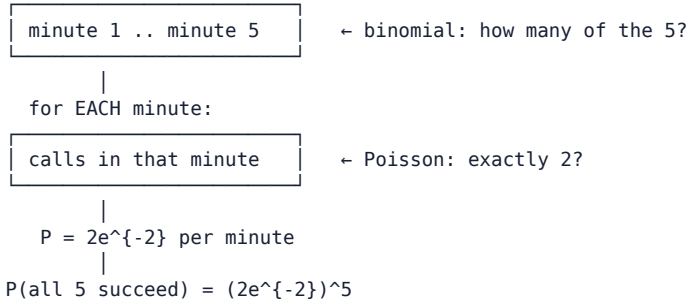
STEP 2: binomial over the FIVE minutes

M = number of minutes (of 5) with exactly 2 calls

$M \sim B(5, 0.270671)$

$$P(M=5) = (2e^{-2})^5 = 32 e^{-10} = 0.00145$$

THE STRUCTURE:



THIS IS THE COMPOSITION (SLOT 9) QUESTION SHAPE.
 The inner model gives a probability. That probability then becomes the p of an outer binomial.
 Recognise: "in EACH of the k intervals" -> nest.

12. WORKED: life insurance (p027)

5000 men aged 42, $P(\text{death in a year}) = 0.001$.

$P(\text{exactly 4 claims})?$

$$\lambda = n p = 5000 \times 0.001 = 5$$

$$P(X=4) = e^{-5} \frac{5^4}{4!} = e^{-5} (625/24) \\ = 0.0067379 \times 26.0417 = 0.175467$$

ERRATA 1: the slide prints 0.1745. The correct value rounds to 0.1755.

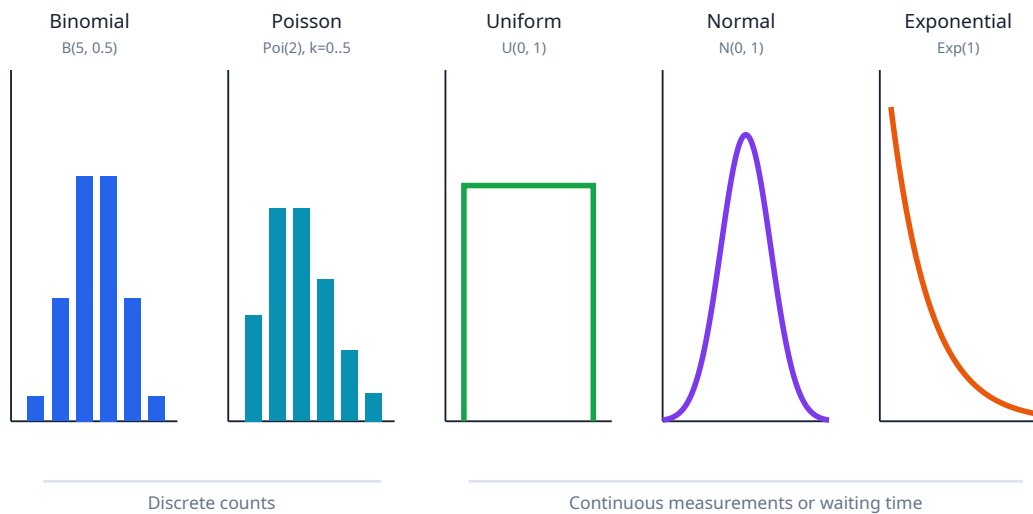
13. The tail form (p028)

$$P(X > 2) = 1 - P(0) - P(1) - P(2)$$

$$\text{general: } P(X > k) = 1 - \sum_{i=0}^k P(X=i)$$

PART C: THE DISCRETE DISTRIBUTIONS SIDE BY SIDE

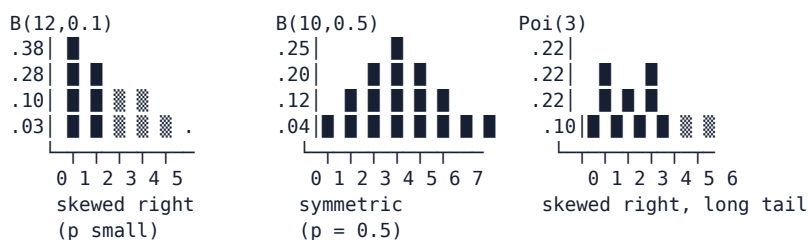
Five distribution shapes



Distribution shapes, with the discrete models on the left

	BINOMIAL	POISSON
models	# successes in n trials	# rare events in a fixed interval
setup	n finite, indep, p constant	rare, indep in time/space
pmf	$C(n,x)p^x q^{n-x}$	$e^{-\lambda} \lambda^x / x!$
mean	np	λ
variance	npq	λ
signature	npq < np always	mean = variance
link	$n \rightarrow \infty, p \rightarrow 0, np = \lambda \rightarrow$	(the limit)

the pmf shapes:



THE RECOGNITION ROUTINE FOR THESE TWO

"n trials, two outcomes, independent, p constant"
 "arrivals, calls, defects, rare, per unit time"
 "use the approximation"
 "in each of the k intervals"
 "mean equals variance"
 "at least one"

-> BINOMIAL
 -> POISSON
 -> POISSON with $\lambda=np$
 -> NEST (Poisson then binomial)
 -> necessary Poisson check;
 confirm the model too
 -> complement, $1 - q^n$